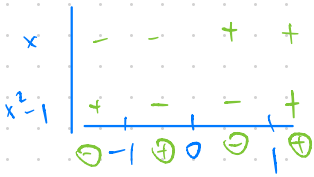
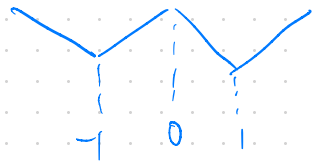
$f(x) = x^4 - 2x^2 + 3$

a) $f'(x) = 4x^3 - 4x$

let $f'(x) = 4x(x^2-1) = 0$

$x = 0$ or $x = \pm 1$





when $x < -1$ or $0 < x < 1$, $f'(x) < 0$, f is decreasing

$-1 < x < 0$ or $x > 1$, $f'(x) > 0$, f is increasing

b) when $x = \pm 1$, $f(x)$ have local minimums

$f(-1) = (-1)^4 - 2 + 3 = 1 - 2 + 3 = 2$

$f(1) = 1^4 - 2 + 3 = 4 - 2 = 2$

when $x = 0$, $f(x)$ has local maximum

$f(0) = 3$

c) $f''(x) = 12x^2 - 4$

when $f''(x) = 4(3x^2 - 1) = 0$

$x = \pm \frac{1}{\sqrt{3}}$

when $x < -\frac{1}{\sqrt{3}}$ or $x > \frac{1}{\sqrt{3}}$, $f(x)$ concave upward,

when $-\frac{1}{\sqrt{3}} < x < \frac{1}{\sqrt{3}}$, $f(x)$ concave downward

$\therefore x = -\frac{1}{\sqrt{3}}$ and $x = \frac{1}{\sqrt{3}}$ are the inflation points



9. $\lim_{x \to 4} \frac{x^2 - 2x - 8}{x - 4}$

$\overset{H}{=} \lim_{x \to 4} \frac{2x - 2}{1}$

$= 6$

15. $\lim_{t \to 0} \frac{e^{2t} - 1}{\sin t} \overset{H}{=} \lim_{t \to 0} \frac{2e^{2t}}{\cos t} = 2$

19. $\lim_{x \to \infty} \frac{\ln x}{\sqrt{x}} \overset{H}{=} \lim_{x \to \infty} \frac{\frac{1}{x}}{\frac{1}{2\sqrt{x}}}$

$= \lim_{x \to \infty} \frac{2\sqrt{x}}{x}$

$= \lim_{x \to \infty} \frac{x^{-\frac{1}{2}}}{1}$

$\overset{H}{=}$

$= 0$

23. $\lim_{t \to 1} \frac{t^8 - 1}{t^5 - 1} \overset{H}{=} \lim_{t \to 1} \frac{8t^7}{5t^4}$

$= \frac{8}{5}$

51. $\lim_{x\to 1}\left(\frac{x}{x-1}-\frac{1}{\ln x}\right)$ $\frac{1}{0}=\infty$ 不定 Indeterminant

$\infty-\infty$ 不定 Indeterminant

$\frac{1}{0}=\infty$

$\overset{H}{=}\lim_{x\to 1}\frac{1}{1}-\lim_{x\to 1}\frac{1}{\frac{1}{x}}$

$=1-1$

$=0$

$\lim_{x\to 1}\frac{x\ln x-x+1}{(x-1)\ln x}$

$\overset{H}{=}\lim_{x\to 1}\frac{1+\ln x-1}{\ln x+\frac{x-1}{x}}$

$=\lim_{x\to 1}\frac{\ln x}{\ln x+1-\frac{1}{x}}$

$\overset{H}{=}\lim_{x\to 1}\frac{\frac{1}{x}}{\frac{1}{x}+x^{-2}}$

$=\frac{1}{2}$

57. $\lim_{x\to 0^+}x^{\sqrt{x}}$

let $y=x^{\sqrt{x}}$

$\ln y=\sqrt{x}\ln x$

$\lim_{x\to 0^+}\ln y=\lim_{x\to 0^+}\sqrt{x}\ln x$

$=\lim_{x\to 0^+}\frac{\ln x}{\frac{1}{\sqrt{x}}}$

$\overset{H}{=}\lim_{x\to 0^+}\frac{\frac{1}{x}}{-\frac{1}{2}x^{-\frac{3}{2}}}$

$=\lim_{x\to 0^+}(-2x^{\frac{1}{2}})$

$=0$

$\lim_{x\to 0^+}y=1$

6s. $\lim_{x\to 0^+} (4x+1)^{\cot x}$

Let $y = (4x+1)^{\cot x}$

$\ln y = \cot x \ln(4x+1)$

$\lim_{x\to 0^+} \ln y = \lim_{x\to 0^+} \cot x \ln(4x+1)$

$= \lim_{x\to 0^+} \dfrac{\ln(4x+1)}{\frac{1}{\cot x}}$

$\overset{H}{=} \lim_{x\to 0^+} \dfrac{\frac{4}{4x+1}}{\sec^2 x}$

$= 4$

$\lim_{x\to 0^+} y = e^4$

56. $\lim_{x \to 1^+} [\ln(x^7 - 1) - \ln(x^5 - 1)]$

$= \lim_{x \to 1^+} \left[\ln \frac{x^7 - 1}{x^5 - 1}\right]$

$= \ln\left[\lim_{x \to 1^+} \frac{x^7 - 1}{x^5 - 1}\right]$ → 存在 of Indeterminant.

$\overset{H}{=} \ln\left[\lim_{x \to 1^+} \frac{7x^6}{5x^4}\right]$

$= \ln\left(\frac{7}{5}\right)$